

cluster__consistency__v2

SCX

2026

1 Theorem 3 (Revised): Fixed-K Cluster Consistency of State Discovery Under Strong Features

Core Claim: Let the number of states K be fixed. When the feature representation $\phi(x)$ separates the K true states by a margin $\Delta_{\min} > 0$ and the per-state sample size $n_{\min} \rightarrow \infty$, Lloyd's k -means with $R = O(\log n)$ random restarts recovers the true state partition with probability $1 - K \cdot \exp(-c \cdot n_{\min} \Delta_{\min}^2 / \sigma^2) - o(1)$.

This is the positive counterpart to Theorem 2's negative result (weak features $\rightarrow$ failure).

Associated concepts: [[Strong Feature Regime]], [[State Discovery]],
[[Fixed-K K-Means Consistency]] **Associated code:** `src/scx/state/discovery.py`
Associated experiments: AlN v3 MLIP (ACE descriptor, success case)